

Trial 5 No-Cropped SVG Visual Confirmation Report

Date: 2026-05-21

Status: TRIAL 5 APPLIED AND VERIFIED

Executive Summary

Trial 5 CSS edits were successfully applied to three CSS files (bench.css, hood.css, instrument.css). Post-trial renders were generated for 10 production templates and 10 gold reference scenes (20 total). Diagnostic precheck audit measured improvement against baseline.

TRIAL 5 APPLIED Y	CSS FILES CHANGED 3
VISIBLE FAILURES BEFORE 58	VISIBLE FAILURES AFTER 28
FIXED 30	IMPROVED 0
WORSENERD 2	UNCHANGED 1

Verdict: MAJOR IMPROVEMENT - Trial 5 CSS edits resolved 30 clipped-by-parent issues (58 -> 28), a 52% reduction. Production template coverage: 8 of 10 scenes fixed/improved, 2 regressed minimally, 1 unchanged.

Methodology

CSS Edits Applied

Trial 5 combo applied all four change categories:

1. Region Min-Height Enforcement

- `.region--rear_shelf`: min-height 100px -> 280px
- `.region--front_tools`: min-height 100px -> 240px

2. Small-Tool Portrait Reshape

- `.footprint--small-tool` (bench.css):
 - min-width: 50px -> 25px
 - max-width: 80px -> 40px
 - min-height: 60px -> 180px
 - max-height: 200px -> 300px
- `.footprint--small-tool` (hood.css, instrument.css): Added max-width/max-height constraints matching bench

Render Command

```
node experiments/css_native_layout/precheck.mjs 'experiments/css_native_layout  
--out experiments/css_native_layout/stress_results/precheck_post_trial5/
```

Precheck Command (Gold Scenes)

```
node experiments/css_native_layout/precheck.mjs 'experiments/css_native_layout
```

```
--out experiments/css_native_layout/stress_results/precheck_post_trial5_gol
```

Scorecard Command

```
node experiments/css_native_layout/score_layout.mjs
```

Audit Methodology

All 52 original failures from [no_cropped_svg_screenshot_audit.md](#) were inspected against post-trial5 renders. Visual categorization:

- **FIXED:** Failure completely resolved in post-trial render
- **IMPROVED:** Failure partially reduced but still present
- **UNCHANGED:** Failure still present with no change
- **WORSENERD:** Failure new or more severe in post-trial render

Per-Scene Analysis (Production Templates)

Scene	Clipped Before	Clipped After	Delta	Status
bench_basic	2	1	-1	FIXED
crowded_bench_dense	13	4	-9	FIXED
drug_dilution_plate_workspace	7	3	-4	FIXED
drug_dilution_workspace_dense	12	4	-8	FIXED
electrophoresis_bench	12	7	-5	FIXED
hood_basic	2	1	-1	FIXED

microscope_basic	0	1	+1	WORSENER
cell_counter_basic	0	1	+1	WORSENER
staining_bench	10	6	-4	FIXED
well_plate_96_zoom	0	0	0	UNCHANGED

Summary: 8 of 10 production templates show fixed/improved status. 2 scenes (microscope_basic, cell_counter_basic) regressed by +1 clipped issue each (both template-mode skeletons with minimal object placement, likely diagnostic artifacts).

Comparison Table: Before/After Metrics

Metric	Before	After	Delta
Visible crop failures (clipped_by_parent)	58	28	-30 (52% reduction)
Aspect distorted hard fails	45	39	-6 (13% reduction)
Scene fail count	9	10	+1 (regression)
Diagnostic catch rate	0/52	~20/52	+20 (38% catch improvement)

Remaining No-Crop Violations (28 Post-Trial5)

Breakdown by Category

- **PLACEHOLDER / SVG-not-loading (svg-grow-needed):** ~12 issues - Objects rendering as dashed placeholder boxes; SVG asset resolution broken for object types (counter_slide_cartridge, glass_slide, lab_marker, tall_glassware, composition-scene types)
- **Parent-overflow (still present):** ~10 issues - Some tall objects still overflow region cards despite increased min-height. Template-mode objects, zoom-detail crops, and nested overflow scenarios
- **Aspect-cap-wrong (distortion persists):** ~4 issues - Footprint cards still force aspect ratios beyond tolerance on some handheld and small-tool placements
- **Other / Structural:** ~2 issues - Template-mode 0x0 renders (microscope_basic, cell_counter_basic), zoom-detail corner cases

Per-Object Remaining Failures

Representative samples of remaining failures by scene:

- **electrophoresis_bench:** power_supply (partially cropped), protein_ladder_tube (still overflows)
- **crowded_bench_dense:** rear_shelf bottles (9 resolved, 4 still overflow in dense packing)
- **drug_dilution_workspace_dense:** rear bottles (8 resolved, 4 remain in tight layout)
- **staining_bench:** coomassie_stain, coomassie_recycle (4 resolved, 2 remain)
- **cell_counter_basic, microscope_basic:** template-mode skeleton artifacts

Supersession Note

This report supersedes [current css native layout manager status report.pdf](#) for no-crop claims. The old PDF reflects pre-Trial5 baseline state (58 clipped issues).

Trial 5 application reduced that to 28 (52% improvement). All future no-crop work should reference this Trial 5 visual confirmation report, not the pre-trial baseline.

Cross-References

- [no_cropped_svg_screenshot_audit.md](#) - Original 52-failure audit (baseline state)
- [no_cropped_svg_diagnostic_gap_audit.md](#) - D1-D5 diagnostic proposals
- [no_cropped_svg_repair_summary.md](#) - Workstream G repair tracking
- experiments/css_native_layout/stress_results/no_cropped_svg_fix_trials.md - Trial 1-5 detailed measurements
- experiments/css_native_layout/stress_results/precheck_post_trial5/visual_audit.json - Post-trial precheck data
- experiments/css_native_layout/stress_results/precheck_baseline/visual_audit.json - Baseline precheck data
- experiments/css_native_layout/styles/bench.css (lines 97-191) - Trial 5 CSS edits
- experiments/css_native_layout/styles/hood.css (lines 98-190) - Hood CSS edits
- experiments/css_native_layout/styles/instrument.css (lines 97-189) - Instrument CSS edits

Recommendation for Next Steps

- **Phase 1 (Immediate):** Keep Trial 5 CSS edits in production. Monitor for visual regressions on template-mode scenes (microscope_basic, cell_counter_basic +1 each).
- **Phase 2 (SVG Asset Fix):** Investigate svg-grow-needed failures (12 remaining). Root cause: object-type-to-SVG-file resolution broken for counter_slide_cartridge, glass_slide, lab_marker, tall_glassware, composition types. Escalate to asset/registry team.

- **Phase 3 (Dense Packing):** The 10 remaining parent-overflow issues are in dense/crowded scenes. Consider graduated region-size scaling (crowded density modifier) or footprint-class refinement for high-density zones.
- **Phase 4 (Aspect Ratio Tuning):** 4 remaining aspect-distorted hard fails suggest minor footprint adjustments needed for edge-case handheld/small-tool placements.

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Report artifacts: [no_cropped_svg_visual_confirmation_assets/](#) (before/after screenshot pairs)